RECOMMENDATION / CONDITIONAL DESIGN CHOICE

Force and useful travel

Alpha=0; 200 mm wheel radius; 325 mm total height including 20 mm buffers at each end.

ROUNDED STRICT OPTION

320 / 320 mm

9:5 guide, original phase zero

USABLE HEIGHT

285 mm

325 total, two 20 mm reserves

WORST FORCE GAIN

1.908 N/Nm

Direct hip input; vertical load

Start with equal 320 mm links and guide ra/rb=108/60=1.8 if the original assembly phase must remain zero. It has a 10.31 degree chord, 9.18 mm maximum bow and 57.83 Nm unassisted static hip peak for an equally loaded 45 kg robot. Normal-branch hip distance stays at least 241.22 mm, 21.22 mm beyond the assumed 220 mm wheel-plus-gap exclusion. Local tangents range 5.05-19.30 degrees;10.31 is the chord. This is a rounded compromise, not the lowest mathematical torque.

Example	Chord deg	Bow mm	Worst F/T N/Nm	Hip peak Nm,45 kg	Hip ROM deg
320/320, 9:5	10.31	9.18	1.908	57.83	38.98
350/350, 7:4	12.02	9.66	1.786	61.80	35.06
425/425, 2% bow	10.05	6.56	1.486	74.25	28.79
300/300, indexed	12.50	0.00	1.927	57.26	44.27

The indexed 300/300 mm option keeps alpha=0, but changes fixed lower-link assembly phase to -25 degrees and uses a 2:1 guide. It produces an exact 12.5 degree line with similar static peak (57.26 Nm), a wider usable hip range (44.27 degrees) and greater speed demand. It is a separately labelled relaxation of the original ratio/length-only request.

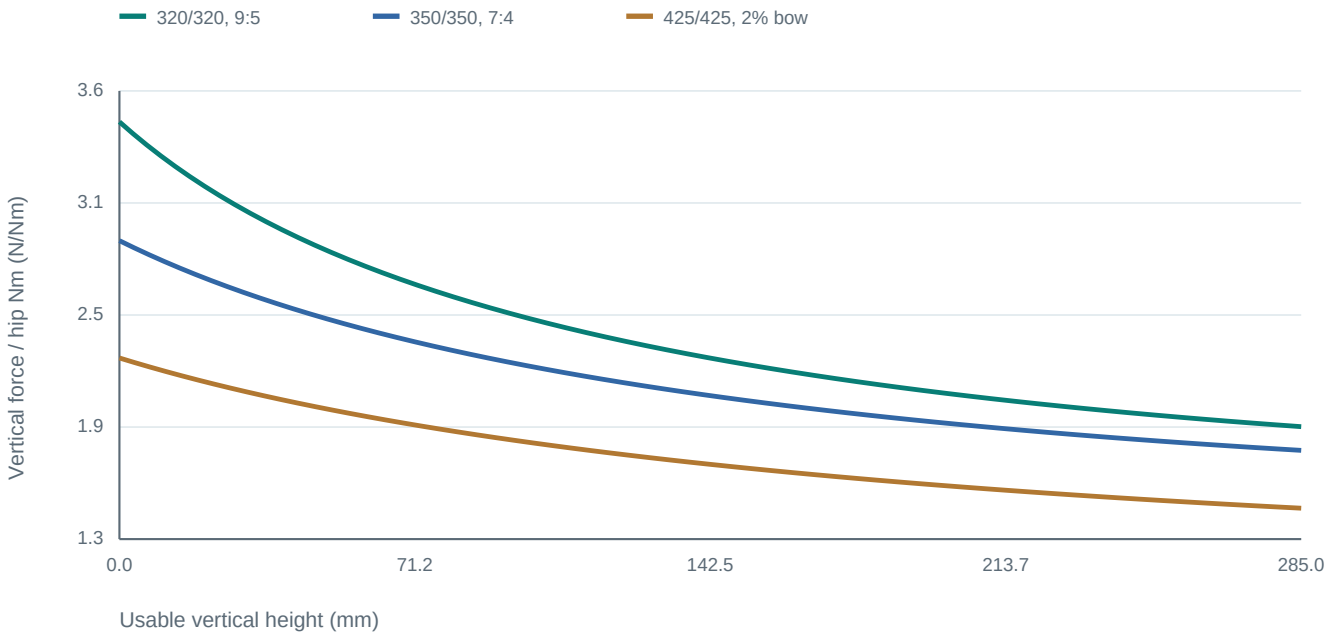
All these legs have an ideal continuous height-inverse through the hip. None has demonstrated physical full inversion: a coplanar 200 mm wheel intersects the hip region at the fold. Axial/chassis packaging, belts, structure and real contact remain unresolved. Historical dual-rope stays IMPOSSIBLE / rejected.

Torque is a transmission tradeoff

q is direct hip-input theta2; Jz=dz/dq, in metres per radian.

```
|tau_hip| = Fz Jz
guaranteed Fz / |tau_hip| = 1 / max(Jz)
RMS_z(tau) = Fz sqrt(integral(Jz^2 dz) / H)
W = integral(tau dq) = Fz H
tau_relative-knee = -tau_hip / r
relative-knee speed = -r hip speed
```

Use the worst force per input Nm for a peak-limited actuator, and height-weighted RMS torque as a secondary measure when moving at constant vertical speed. Angle-average torque is Fz H / ROM: a larger joint sweep can lower it without improving energy efficiency. RMS requires time weighting once the real cycle includes acceleration or dwell.



The ideal 285 mm lift needs 27.96-31.45 J per leg for 40-45 kg four-leg static loading, regardless of these ratios. Lower torque trades for a larger/faster joint motion. The knee coordinate changes torque and speed together; an additional motor gearbox is a separate mapping. No electrical efficiency or thermal rating is inferred.

SEARCH / REPRODUCIBILITY AND ACTIVE BOUNDS

Grid first, local refinement second

11,478,060 strict combinations; 11,070 separate fixed-indexing combinations.

Input / constraint	Declared policy
L3 / L2:L3	200-450 mm; 30 scales. Ratio 0.85-1.15 step 0.01; equal/5/10/15% tiers
Guide / start	r1.4-2.6 step 0.005 plus exact 5/3; q100-140 step 2.5; end<=175
Travel / reserves	300/325/350 mm total; actual inverse-height 20 mm trim at both ends
Path quality	Chord 10-15 deg; bow<=3%chord; tangent span<=15 deg. 2% compared separately
Feasibility	Normal |C|>=220 mm; full q100-260 Jz>=5 mm/rad; L2>60(r+1)
Ranking / sampling	Minimum useful-stroke maxJz, then size; 51 coarse /2001 finalist poses
Second method	SLSQP seeded by grid; two 325 mm refinements, 401-pose constraints; dense check

Locally refined	Links mm	Guide	Hip peak Nm	Worst F/T
303/303, refined	303.46/303.46	1.807104	55.423	1.9913
294/309, refined	293.78/309.24	1.786363	54.206	2.0360

The equal local optimum is approximately 303.46/303.46 mm; allowing 5% mismatch gives 293.78/309.24 mm. Both sit at chord 10 degrees, tangent span 15 degrees and the 220 mm clearance boundary. Their torque improvements are conditional and leave no surplus beyond that assumed gap. The near-equal result also reaches its 5% mismatch bound. The rounded 320 mm case adds useful packaging allowance. Local convergence and bounded grids do not establish a global or hardware optimum.

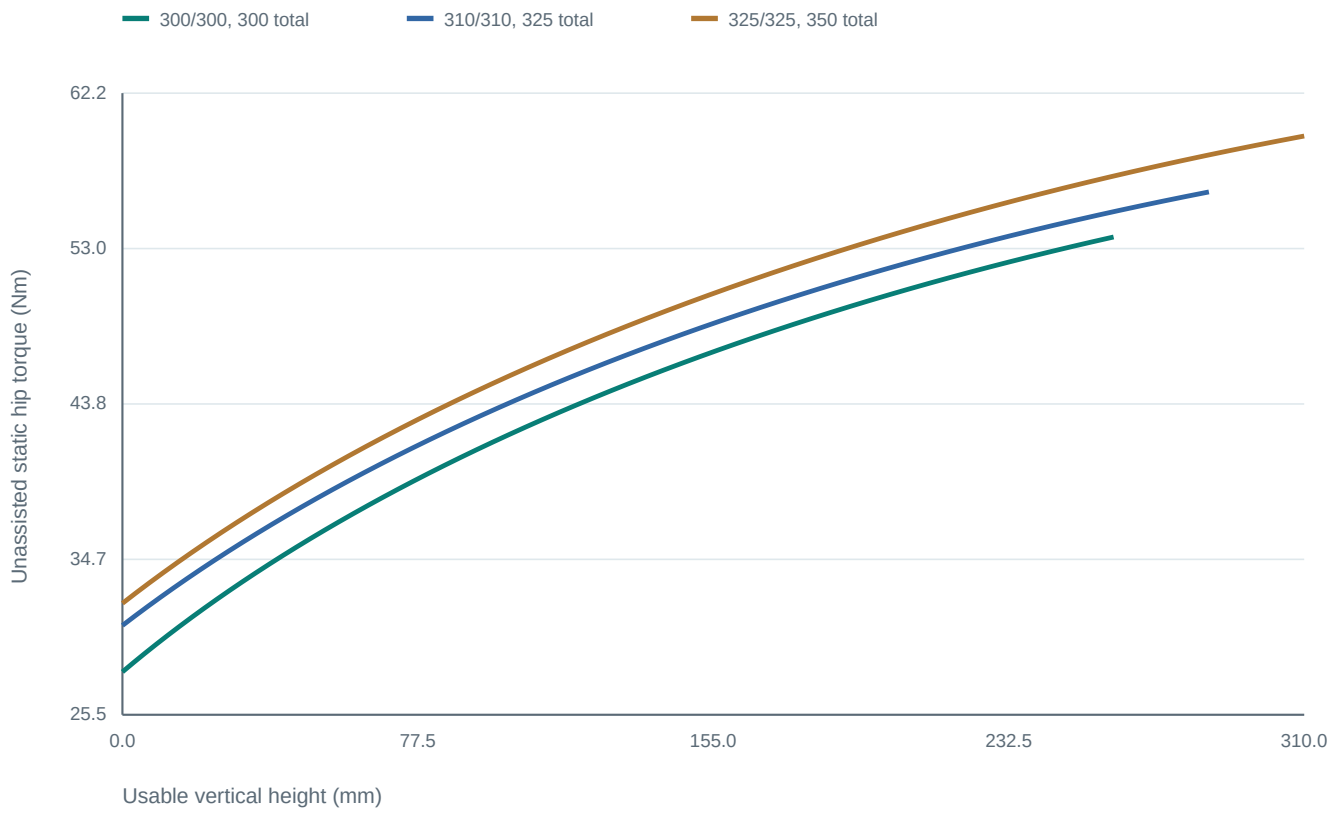
Quality counts use coarse samples. All published candidate acceptance and profile metrics are densely rechecked. The first evidence export rejected NumPy boolean serialization; explicit scalar conversion fixed export and the complete search was rerun.

TRAVEL / CAPACITY VERSUS LEVERAGE

300, 325 or 350 mm total?

Equal-link phase-zero grid winners under the same 3%bow /15 degree-span policy.

L2=L3 mm	Guide	Total / usable mm	Hip ROM deg	Hip peak Nm,45 kg	Worst F/T N/Nm
300	1.810	300/260	38.93	53.68	2.056
310	1.795	325/285	39.98	56.33	1.959
325	1.800	350/310	41.20	59.63	1.851

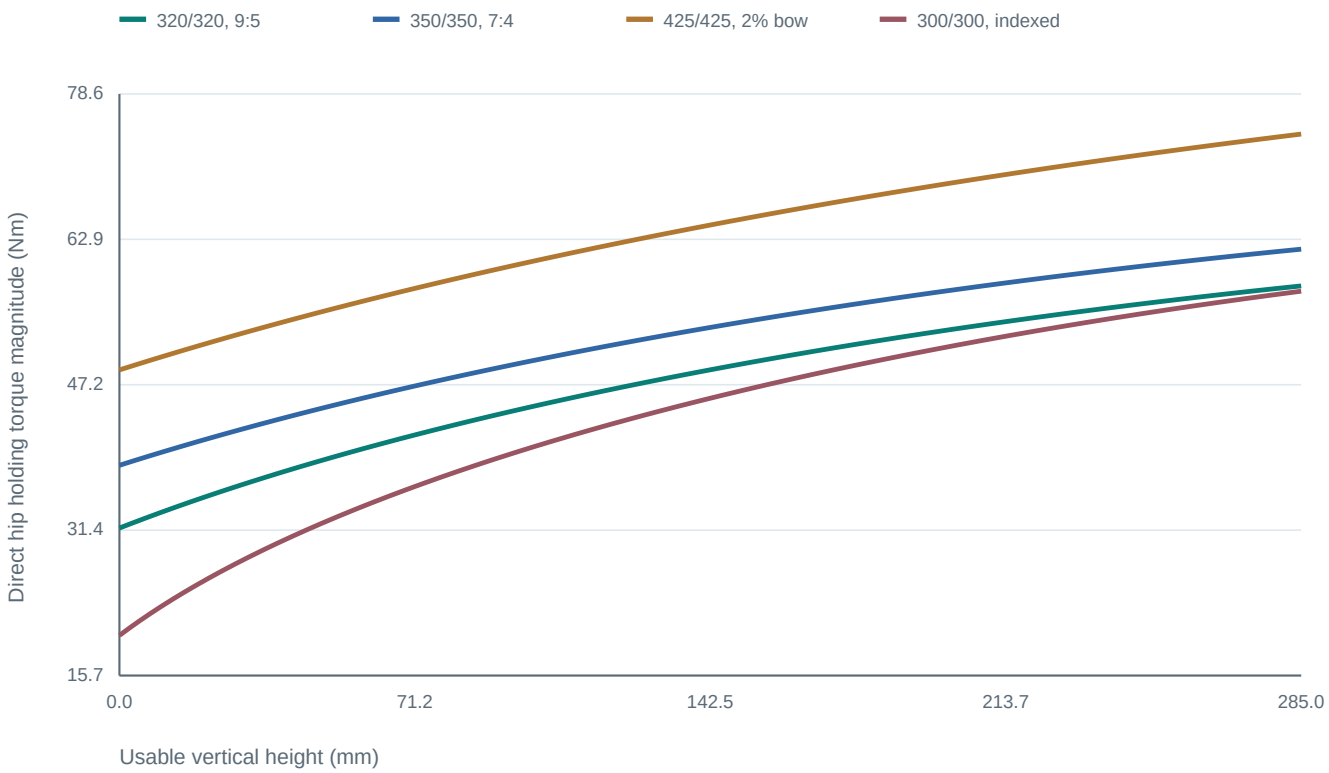


300 mm total gives 260 mm usable and already exceeds the owner's approximate 250 mm preferred useful height. Choose 325 mm as a middle study point, or 350 mm if the extra 50 mm useful height compared with 300 is valuable. Extra travel requires more size or joint motion and generally weakens worst-case force leverage here. Buffers are geometric end reserves, not a measured impact stroke or spring deflection allowance.

INDIVIDUAL CURVES / SAME TRAVEL AND LOAD

The force-oriented shortlist

All four shown cases: 325 mm total /285 mm usable, Fy=0, 11.25 kg equivalent static corner load.



Design	40 kg /1 g hip peak	45 kg /1 g hip peak	45 kg /3 g hip peak	45 kg /4 g hip peak
320/320, 9:5	51.4	57.8	173.5	231.3
350/350, 7:4	54.9	61.8	185.4	247.2
425/425, 2% bow	66.0	74.2	222.7	297.0
300/300, indexed	50.9	57.3	171.8	229.1

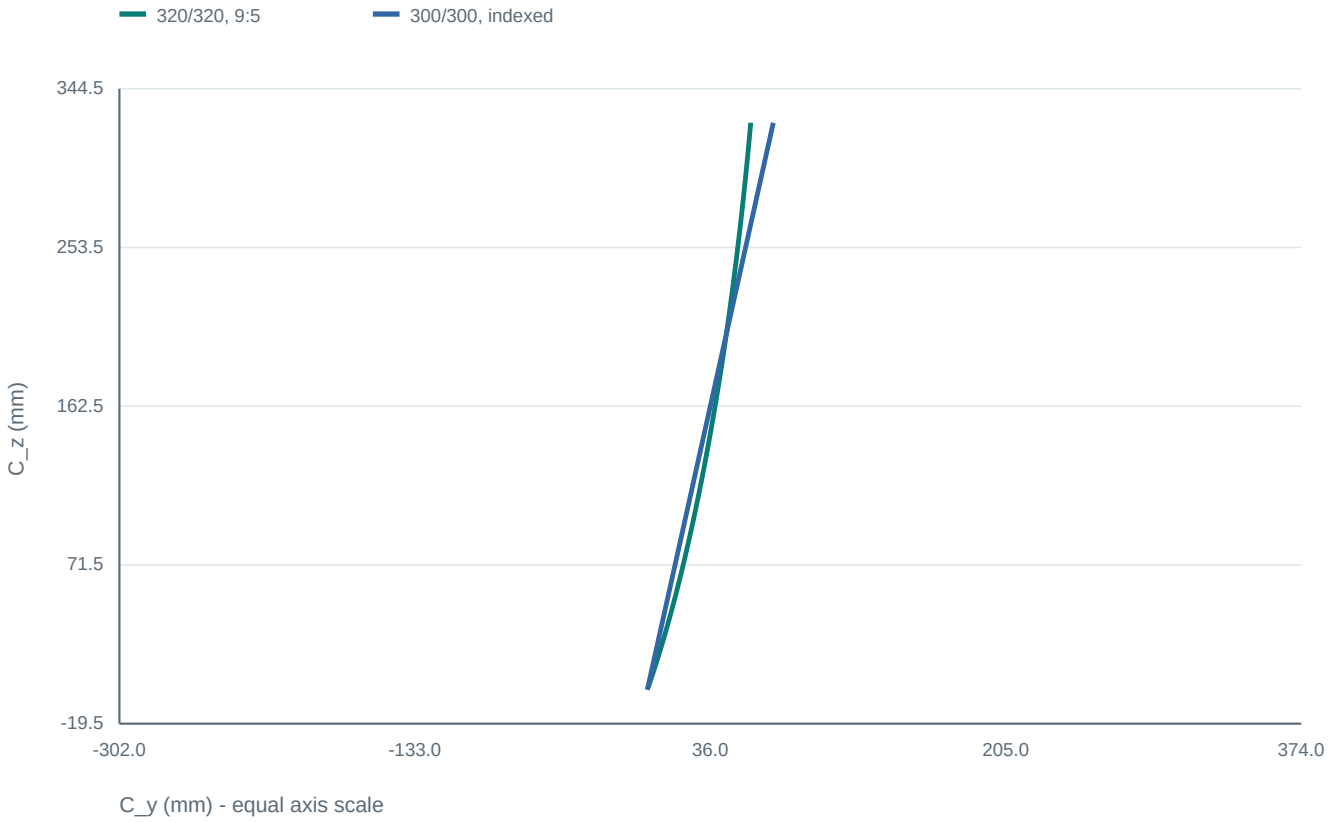
These numbers are unassisted equivalent-load holding torques. 3 g/4 g scales the total supported force, rather than adding 3 g/4 g on top of gravity. They do not include motor/gear inertia, link masses, transient impacts or spring assistance. Do not transfer the earlier unequal-leg spring or 6.449 Nm dynamic example to these geometries.

INDEXING / SEPARATE DESIGN RELAXATION

An exact tilted line at alpha zero

Fixed lower-link indexing phi0=-25 degrees, guide r=2; no ratio change or reindexing during movement.

For equal links and d=phi0/2:
`C=2L sin(q-pi-d) [-sin(d),cos(d)]`
The wheel centre stays on an exact 12.5 degree line. Original phase zero cannot achieve this same tilted line through inversion because x is even and z odd aboutq180 degrees.



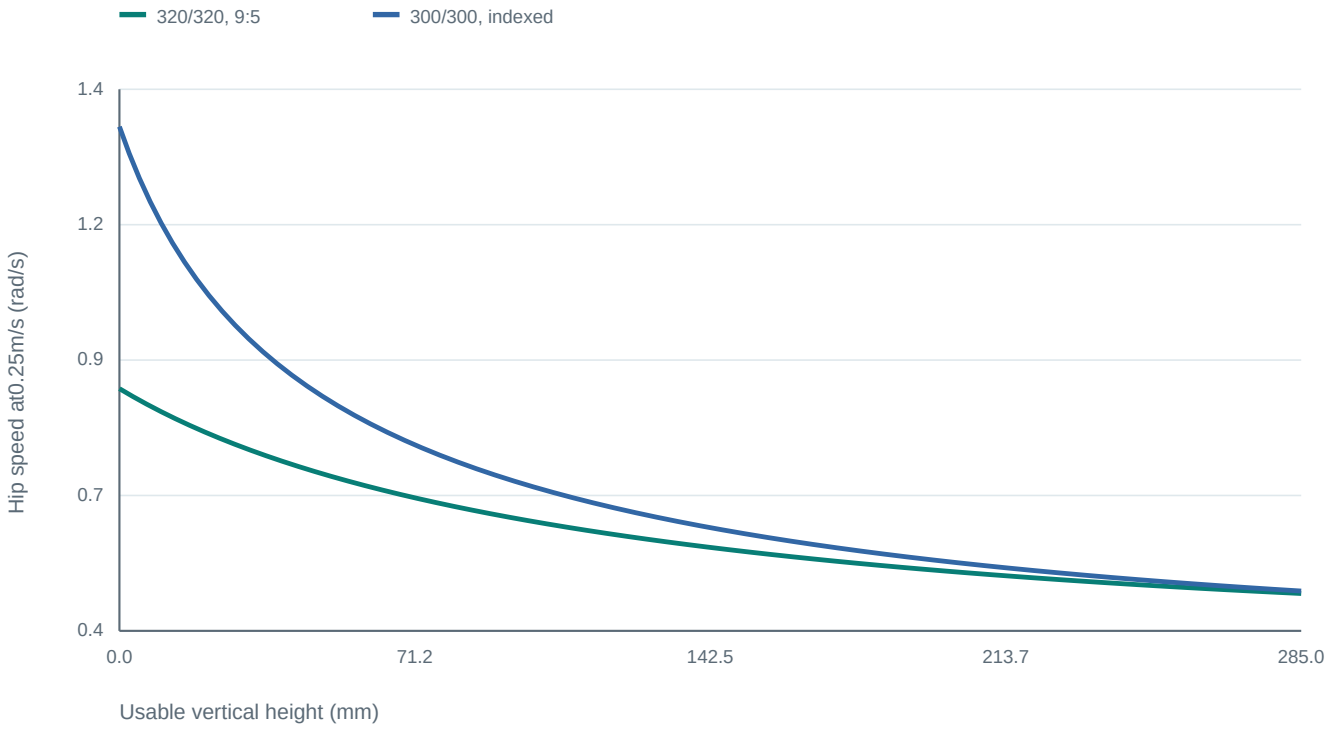
Parameter	Strict rounded	Indexed rounded
Links / guide	320/320 mm;9:5	300/300 mm;2:1
Total angle window	110-155.398 deg	87.5-142.033 deg
Useful hip range	38.98 deg	44.27 deg
Useful peak at 0.25m/s	0.871 rad/s	1.376 rad/s

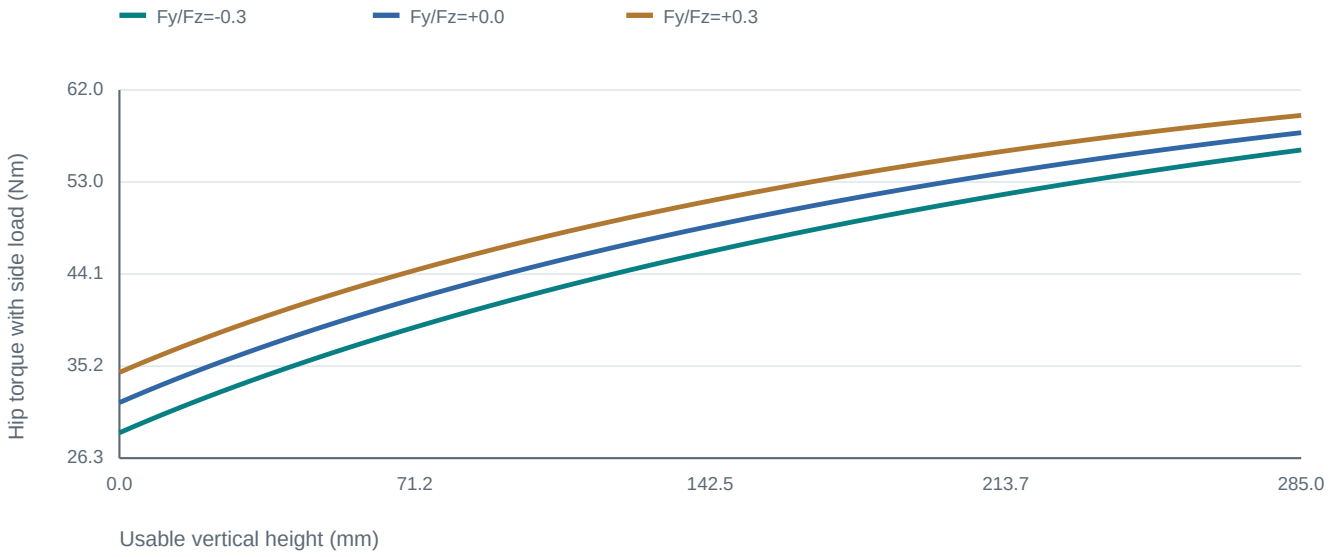
Indexed 285 mm is closer to clearance; rounded 300 mm adds margin. This family needs the shifted 87.5-247.5 degree envelope, outside the old V2 validator. Original app validators and live presets are unchanged.

CONTROL / FORCE, VELOCITY AND SIDE LOAD

Check more than average torque

Illustrative constant vertical speed 0.25 m/s; not a smooth-start actuator specification.



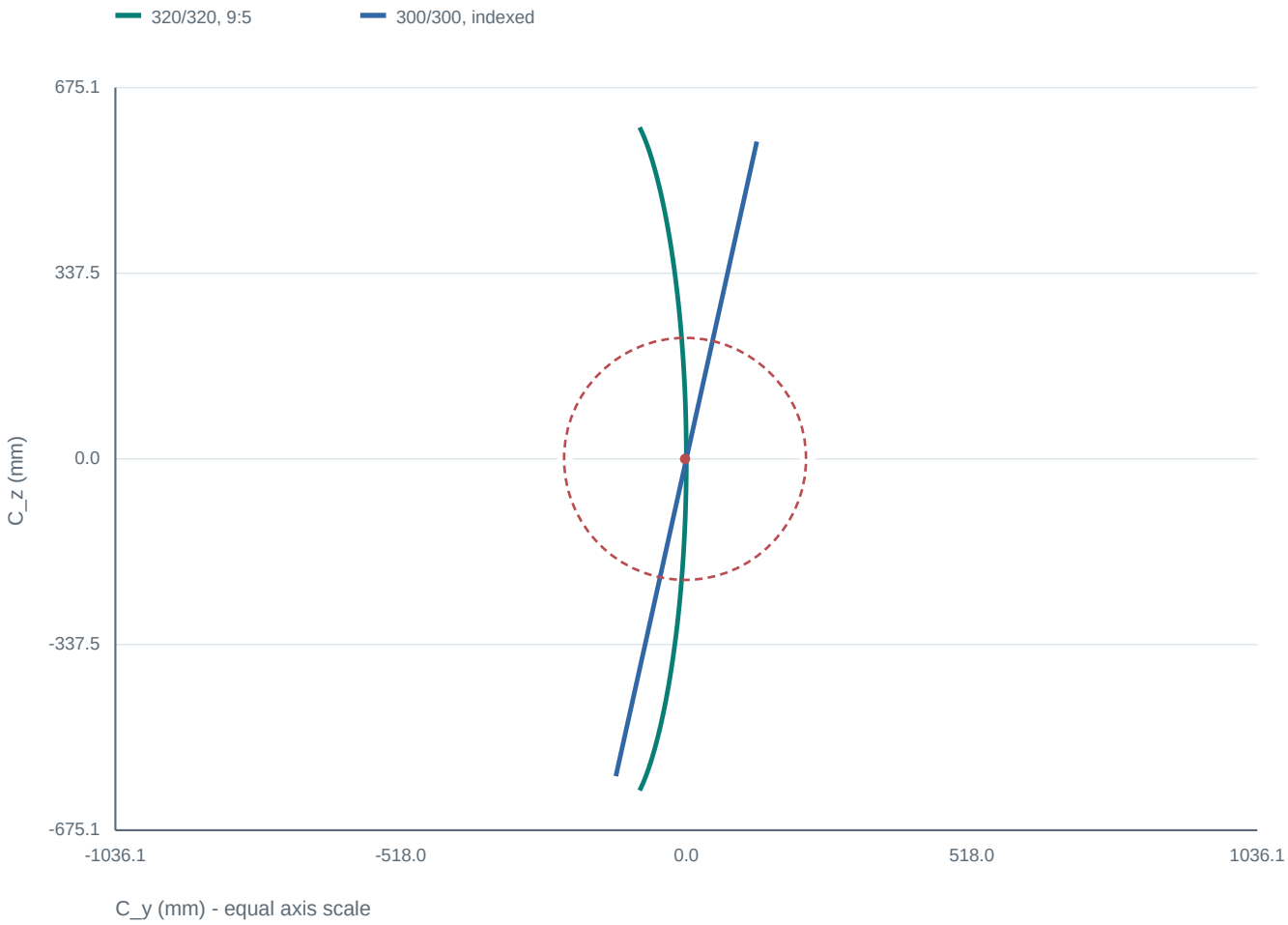


General endpoint load uses tau=-(Fy Jx+Fz Jz). A side load can raise or reduce the instantaneous demand; the pure-vertical ranking is not universal. Smooth motion also needs qddot=zddot/Jz-(dJz/dq)zdot^2/Jz^3. Actual acceleration/duty, moving masses, spring geometry, damping and impact/contact data are needed before selecting a motor.

INVERSION / MATHEMATICAL AND PHYSICAL CHECKS

Ideal invertibility is conditional

Neither a positive Jacobian nor a collision-disabled Pymunk sweep proves wheel/chassis clearance.



Actual Pymunk case	2 ms error mm	1 ms error mm	0.5 ms error mm
320/320, 9:5	0.11426	0.05673	0.02826
294/309, refined	0.10465	0.05196	0.02589
300/300, indexed	0.13657	0.06780	0.03378

All three full angular sweeps pass the unchanged 0.1 mm path/pin thresholds at 1 and 0.5 ms; every 2 ms path result fails and is retained. At the fold, near-equal link distance is |L2-L3|, well below the 220 mm exclusion. An axial stagger may change collision geometry, but its dimensions are unknown. The strict branch mirrors after inversion and has a vertical fold tangent; the indexed line keeps its tilt. These engine runs check positive-mass ideal constraints with no gravity, contact or collision shapes.

VALIDATION / SOURCES AND NEXT DECISIONS

Use the ranking as a design screen

Preserve the old reports, models, Motion Lab illustrations and historical decisions.

Evidence	Result / boundary
Numerical tests	5 new force tests; existing 5 inversion and 12 generalized tests pass
Original V2 + Andy statics	10,248 scalar comparisons; max error 3.41e-13 mm/Nm
Actual constrained Pymunk	3 new full-sweep refinements; geometry only, no reset after initialization
Independent math review	Force/work/knee mapping, analytic peaks and selected dense path checks
Offline delivery	Static HTML with embedded SVG/JSON, local PDF, no server or connection
Canonical / hardware	Not applicable /not certified; no strength, actual impact or packaging pass

Practical choice: use 320/320 mm, 9:5 when fixed phase zero is required and 9 mm bow is acceptable. Keep 350/350 mm, 7:4 if a 12 degree chord and existing example are preferred. Use 425 mm tested links only if tighter 2%bow dominates force and size. If fixed assembly indexing is allowed,300/300 mm, 2:1 with phi0=-25 gives the cleanest line, with a wider control/speed range. These are recommendations within this declared ideal domain.

Next measure wheel/chassis axial separation and physical sweep clearance; agree accepted local tangent/bow limits, motor-shaft mapping and actual usable impact stroke. Then allocate measured moving masses, refit the chosen spring topology and run a smooth supported-chassis contact model. A different load cycle can change the best RMS design.

Sources: Jonathan leg-solver V2 embedded solver/statics; existing Andy model.js and generalized_leg.py; retained force-travel search, tests, math and engine evidence. Commands and equations are in docs/force-travel-study.md. Base d1596e5; date 2026-10-14. The screenshot's Fy=Fz=100 N case remains separate from this pure-vertical 40-45 kg load comparison. No new motor/spring product or canonical geometry is published.